AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	NAO 1/NAO 2, OH -- station KCMH, America/New_York
Committed pair	OSM 671301768 -> 802160675 NAO 1/NAO 2 / NAO 3
Facade gap	115.9 m between the two halls
Weather record	43,786 real hourly records from KCMH
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.4443 C at 300 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2024-02-28 (crossing)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 8 of 24 hours with 1 mode change. That is 3 more than the reactive incumbent, at 0 unsafe hours against its 2. 5 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	8 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	5 of 24 hours free cooling, 2 mode change(s), 2 unsafe hour(s)
	5 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	dew point	15.809	18.0	16.110	4.078
01	mechanical	no	dew point	15.805	18.0	16.110	3.654
02	mechanical	no	dew point	16.319	18.0	16.670	3.646
03	mechanical	yes	switch budget	16.403	18.0	18.330	3.300
04	mechanical	yes	switch budget	16.319	18.0	18.890	2.873
05	mechanical	yes	switch budget	16.285	18.0	17.220	2.477
06	mechanical	yes	switch budget	16.951	18.0	15.002	2.679

07	mechanical	yes	switch budget	17.996	18.0	15.000	2.361
08	mechanical	no	dry-bulb	20.194	18.0	14.440	2.792
09	mechanical	no	dry-bulb	24.686	18.0	10.853	3.863
10	mechanical	no	dry-bulb	28.431	18.0	6.823	4.999
11	mechanical	no	dry-bulb	28.950	18.0	4.593	5.850
12	mechanical	no	dry-bulb	27.770	18.0	3.459	5.988
13	mechanical	no	dry-bulb	27.432	18.0	2.782	5.782
14	mechanical	no	dry-bulb	25.184	18.0	1.672	4.861
15	mechanical	no	dry-bulb	19.592	18.0	1.672	4.428
16	FREE	yes	-	13.354	18.0	0.689	3.351
17	FREE	yes	-	9.160	18.0	0.153	3.375
18	FREE	yes	-	6.182	18.0	-0.707	2.793
19	FREE	yes	-	4.642	18.0	-1.237	3.208
20	FREE	yes	-	2.156	18.0	-1.787	3.490
21	FREE	yes	-	1.768	18.0	-2.347	3.839
22	FREE	yes	-	0.709	18.0	-3.465	4.289
23	FREE	yes	-	-0.097	18.0	-3.796	4.328

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 15.809 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

01:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 15.805 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

02:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 16.319 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.403 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.319 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.285 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.952 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.996 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.194 C against a 18.0 C limit, so it fails by 2.194 C. It would take a limit 2.194 C higher, or a bound 2.194 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.687 C against a 18.0 C limit, so it fails by 6.687 C. It would take a limit 6.687 C higher, or a bound 6.687 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.431 C against a 18.0 C limit, so it fails by 10.431 C. It would take a limit 10.431 C higher, or a bound 10.431 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.949 C against a 18.0 C limit, so it fails by 10.949 C. It would take a limit 10.949 C higher, or a bound 10.949 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.770 C against a 18.0 C limit, so it fails by 9.770 C. It would take a limit 9.770 C higher, or a bound 9.770 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.432 C against a 18.0 C limit, so it fails by 9.432 C. It would take a limit 9.432 C higher, or a bound 9.432 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.184 C against a 18.0 C limit, so it fails by 7.184 C. It would take a limit 7.184 C higher, or a bound 7.184 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.592 C against a 18.0 C limit, so it fails by 1.592 C. It would take a limit 1.592 C higher, or a bound 1.592 C tighter, to change this hour.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.354 C, which is 4.646 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.351 C, and the margin is measured, not chosen: 3.102 C of group-conditional forecast error for this hour of day, plus 0.2483 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.160 C, which is 8.840 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.375 C, and the margin is measured, not chosen: 3.096 C of group-conditional forecast error for this hour of day, plus 0.2789 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.182 C, which is 11.818 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.793 C, and the margin is measured, not chosen: 2.494 C of group-conditional forecast error for this hour of day, plus 0.2990 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 4.642 C, which is 13.358 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.208 C, and the margin is measured, not chosen: 2.880 C of group-conditional forecast error for this hour of day, plus 0.3279 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 2.156 C, which is 15.844 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.490 C, and the margin is measured, not chosen: 3.181 C of group-conditional forecast error for this hour of day, plus 0.3094 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 1.768 C, which is 16.232 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.839 C, and the margin is measured, not chosen: 3.527 C of group-conditional forecast error for this hour of day, plus 0.3127 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.709 C, which is 17.291 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.289 C, and the margin is measured, not chosen: 3.973 C of group-conditional forecast error for this hour of day, plus 0.3163 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -0.097 C, which is 18.097 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.328 C, and the margin is measured, not chosen: 4.027 C of group-conditional forecast error for this hour of day, plus 0.3009 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

Generated by AGENTIC-ARBITER/src/report.py. Verified by being read back: the file was reopened after writing and every hour of the schedule, the headline counts and this site's own name were confirmed present.